2 System wiring overview

Every assembly on one page, joined by real wires:
battery to motors, the safety chain, rails, ESP32,
I2C chain. Drawing only; nets are on pages 3–9.

File: overview.kicad_sch

3 Car harness and box edge

Everything outside the box: battery + F1, key, pedal,
shifter, 5 motors, E–stop, sensors, dash, speakers.
Box plugs J1–J12 (J1–J8 mate the stock plugs, TBD #17).

File: harness.kicad_sch

4 Power and safety chain

K1 contactor and its coil chain: key, E–stop (2 NC),
Q1 hold MOSFET. F2 key feed, fuse block F3–F8,
ground star TB1, motor bus TB2, battery divider,
motor–bus sense and bleed.

File: power.kicad_sch

5 Low–voltage rails

U1 LM2596 5V_LOGIC, U2 AMS1117 3V3_SENSE,
U3 5V_ACC and U4 12V_ACC for accessories.

File: rails.kicad_sch

6 Drive, variant A (4 + 1 IBT–2)

U10–U13 IBT–2 per wheel, U14 steering IBT–2.
470 µF each, enable pull–down per board,
IS current–sense networks to the ADS1115s.

File: drive.kicad_sch

7 Controller and inputs

ESP32 (nets from include/pins.h), ELRS receiver,
pedal input protection, wheel–speed pull–ups,
DS18B20, DFPlayer serial, listen–only Bluetooth log.

File: controller.kicad_sch

8 I2C bus and switch inputs

2 x ADS1115 current sense, PCF8574 switches
(shifter, horn, E–stop sense, seat), MPU–6050,
PCA9685 accessory PWM.

File: i2c_io.kicad_sch

9 Accessories: fans, lights, sound

MOSFET board for lights and fans, 12 V branch
fuses F9/F10, music amp, DFPlayer + alert speaker.

File: accessories.kicad_sch

How to read it

Every pin has a stub and a net label: the label names the net, so each pin says where it goes. Labels with a flag shape are global (used on more than one sheet); plain text labels stay on one sheet. Each box above is a sheet: double–click it in KiCad to open it (or page down in the PDF). Location field: box, A1 (the interface board), car, dash. Source field: owned, ordered 24 Oct, stock car, to buy, TBD. Tables: NETS.md, CONNECTORS.md, WIRE_LIST.md, BOM.md (same data).

Three safety layers (POWER_CHAIN.md)

1. A power chain a person can always cut, no software:
   F1, key SW1 and E–stop S2 (two NC blocks) in the K1 coil.
2. Drivers that default off: DRIVE_EN / STEER_EN pulled down
   at every board; K1 coil MOSFET Q1 pulled down.
3. The ESP32 only adds control on top, and can only open K1.

Open items (TBD)

#17 stock plug types and pin use: J1–J8 (car–facing plugs)
#33 motor pairing in the harness: J6/J7 splitter or not
#13 battery voltage and chemistry: D2 TVS value, BTS7960 27.5 V limit
#14 motor labels: per–motor current, variant A or B (#32)
#18 box size: variant A (5 boards) or B (2 boards)
#16 pedal type: switch, pot or 5 V hall (divider)
#15 IBT–2 IS resistor: config IS_RESISTOR_OHMS
Key switch rating (SW1 carries ~8 A worst case)
Shifter: signal–only or motor current (only signal can go to U24)
K1 precharge resistor: after the bench test
AS5600 cable length: I2C or analog OUT on U23 A1
Variant B current sensors (#32)

Revisions

0.3 2026–10–24 First KiCad version, from draft v0.2 (SVG).
Net names, connector/pin tables, wire list, BOM
matched to the 24 Oct order. Changes are listed
in README.md, 'Changes from draft v0.2'.

Draft v0.3, not for building the car–facing side until the TBDs close

RideOnESP32 · Can–Am Maverick X3 24 V 4WD

Sheet: /
File: rideonesp32.kicad_sch

Title: RideOnESP32 v3 wiring schematic (index)

Size: A3	Date: 2026–10–24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 1/9

System wiring overview: everything connected

Each thin line is one wire; thick blue, green and grey lines are labelled bundles. Wire numbers, gauges and every single conductor are in WIRE_LIST.md; pins in CONNECTORS.md. Drawing only: the nets live on sheets 3–9.

CAR SIDE (stock parts and our off-box parts)

BT1 battery
24 V, chemistry TBD (#13)
+
−

W001 10 AWG

F1 40 A
mr15 cm from + b
a
b

J1 BATT
+
−

10 AWG

K1 contactor
125 A, 24 V coil
A
C+
C−

F2 15 A
inline
a
b

J2 KEY
1
2

SW1 key
stock dash
1
2

key_24V

FB1 fuse block
bus = KEY_24V
IN
F3
F4
F5
F6
F7
F8

F3 → K1 coil

J9 E-STOP
1
2

S2 E-STOP
latching, 2 NC in series + 1 NO
NC in
NC out
NO
NO gnd

J3 PEDAL
1
2
3

Pedal
type TBD (#16)
V+
SIG
GND

J4 SHIFT
1
2
3

Shifter F/R
stock, signal only
F
R
GND

J8 ACC
1
2
3
4

Horn button, lights
stock (TBD #17)
L+
L−
HORN
GND

J10 SENS
1
2
3
4
5
6
7
8

Sensors
H1/H2 A3144 at the wheels, U25 AS5600 on the steering column, SW5 seat
5V
GND
WS_L
WS_R
3V3
SCL
SDA
SEAT

J11 DASH
1
2
3
4
5
6

Dash
U27 OLED, LS1 alert speaker
3V3
GND
SCL
SDA
SPK+
SPK−

J12 MUSIC
1
2
3
4

Music speakers
LS2, LS3 (+ tweeters)
L+
L−
R+
R−

J12 = TB4 GND star

THE BOX (in the stock controller's place under the seat)

U2 AMS1117 3.3 V
3V3_SENSE
out
in

U1 LM2596 5 V
5V_LOGIC
in
out
gnd

U3 buck 5 V 5 A
5V_ACC
in
out
gnd

U4 buck 12 V
12V_ACC
in
out
gnd

U20 ESP32
pins from pins.h
VIN/GND
analog
GPIO12
FL
FR
RL
RR
SCL/SDA
RX/TX
TX0
S7

A1 interface board
Q1, IS RC + clamps, pedal and hall inputs, divider, buses
Q1 drain
pedal
halls
buses out
to ESP32
5V/3V3 out
IS in
IS out
GND
5V in
3V3 in

U21 ELRS receiver
CRSF, antenna outside
5V
UART

F9 / F10
3 A lights, 10 A amp
in
ilights
amp

U10 IBT-2 FL
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

U11 IBT-2 FR
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

U12 IBT-2 RL
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

U13 IBT-2 RR
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

U14 IBT-2 STEER
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

FAN1/2 24 V
4box fans
−

U29 MOSFET board
low-side switches
in
fan
ilights

U30 DFPlayer
car sounds
pwr
spk

U33 Bluetooth log
HM-10 BLE, listen-only
in

U32 music amp
ZK-502T, its own Bluetooth
in
out

U22
ADS1115 0x48
in
out
IS x5

U23
ADS1115 0x49
in
out

U24
PCF8574 0x20
in
out
P0–P7

U26
MPU-6050
in
out

U28
PCA9685
in
out
PWM

K1 → TB2, 10 AWG

TB2 MOTOR_24V

IPWM, DPWM, EN, SIS (QED)

IS_FL…RR

CONTACTOR

MOSFET gate, analog

3V3

F8

5V_ACC

12V_ACC

SCL, SDA

I2C to J10 / J11

I2C bus

PWM

UART

IS filtered

SOUND_TX

USB 5V

12 V lights

lights low side

5V_ACC

12V_ACC

F1 → fuse

A1 ground bus → TB1

alert speaker

music 0/A

TB1 GND star

MOTORS (stock)

J6 FRONT
1
2
3
4

M1 FL
C 100 nF on the tabs
+
−

M2 FR
C 100 nF on the tabs
+
−

J7 REAR
1
2
3
4

M3 RL
C 100 nF on the tabs
+
−

M4 RR
C 100 nF on the tabs
+
−

J6/J7: one pair per motor. If the stock harness pairs motors, a reversible splitter at the motor plugs (#33).

J5 STEER
1
2
3
4

M5 steering
C 100 nF on the tabs
+
−

Legend

- Battery + before K1, 10 AWG
- MOTOR_24V after K1, 10 AWG
- Power ground to TB1, 10 AWG
- Motor leads, 12 AWG
- Key-switched 24 V branches (F2–F8)
- K1 coil chain (F3 1 A)
- 5 V
- 3.3 V
- 12 V accessories
- Driver control / PWM bundles
- I2C bus
- Analog: IS, pedal
- Switches, sensors, serial
- Logic ground

Draft v0.3, not for building the car-facing side until the TBDs close

RideOnESP32 · Can-Am Maverick X3 24 V 4WD

Sheet: /2 System wiring overview/
File: overview.kicad_sch

Title: System wiring overview

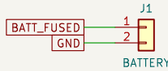
Size: A2	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 2/9

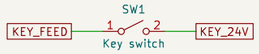
Left: what stays in the car (stock parts and our off-box parts). Right of each plug: into the box.
Plugs facing the stock harness (J1–J8) are TBD until the stock plug photos (#17).
J9–J12 are our own sealed plugs.

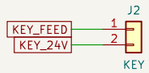
Battery and key



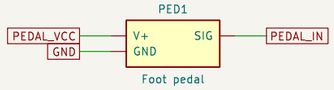


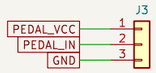






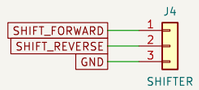
Pedal and shifter



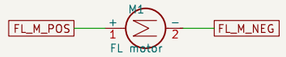








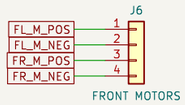
Front motors



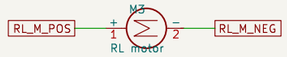






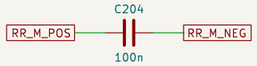


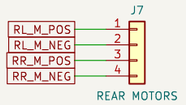
Rear motors







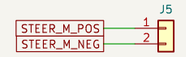


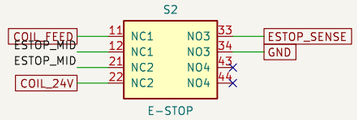


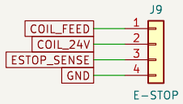
Steering motor, E-stop



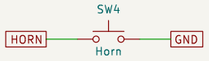




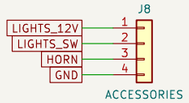




Horn and lights





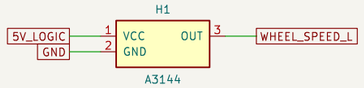


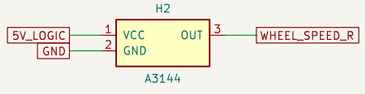
Going back to stock

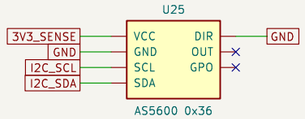
1. Key off. Unplug J1–J8 from the box.
2. Plug the stock HY2005RX controller into the same plugs (motor splitters: unplug, rejoin).
3. Unplug J9–J12; the stock car has none of them.

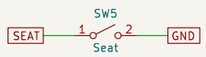
No stock wire is ever cut (#33).

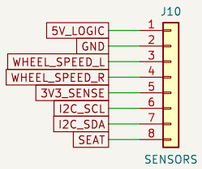
Sensors



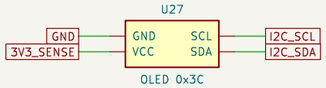


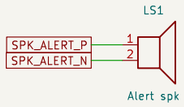


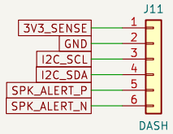




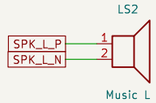
Dash

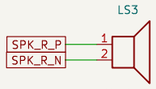


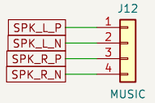




Music







Draft v0.3, not for building the car-facing side until the TBDs close

RideOnESP32 · Can-Am Maverick X3 24 V 4WD

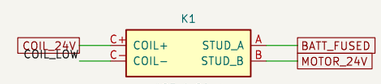
Sheet: /3 Car harness and box edge/
File: harness.kicad_sch

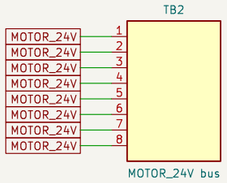
Title: Car harness and box edge

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 3/9

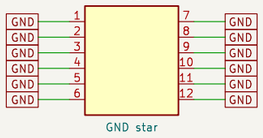
Layer 1, no software: key and E-stop in series with the K1 coil; either one opens K1 whatever the ESP32 does.
Q1 lets the ESP32 open K1 but never close it. Reset, crash or brown-out opens it (R2 pull-down).
Rails come from the key side, before K1, so the ESP32 stays up after an E-stop and can report why.
R5-R7: the ESP32 reads MOTOR_24V on U23 A2. At boot K1 stays open until R7 has drained the bus; still live = K1 welded.

Main path

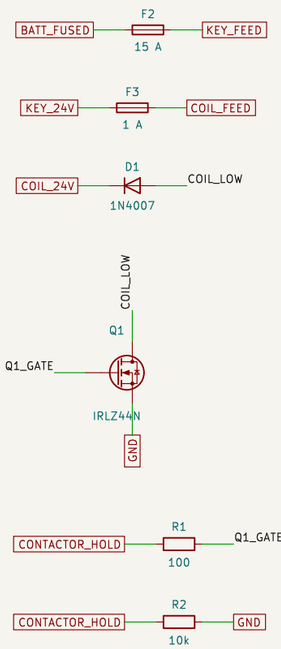








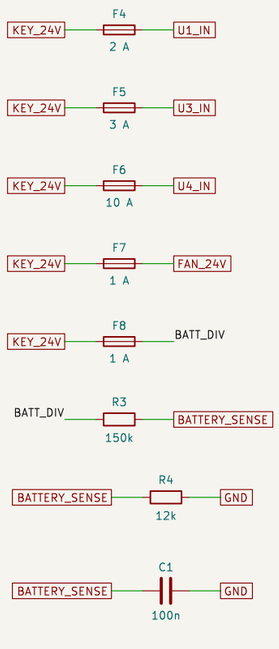
Coil chain



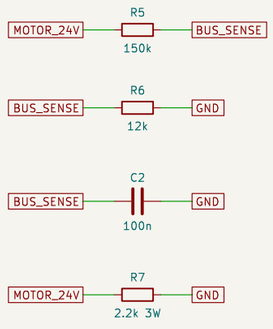
Coil chain

KEY_24V -> F3 1 A -> J9.1 -> S2 NC1 -> S2 NC2
-> J9.2 -> K1 coil -> Q1 -> GND.
Unplugging J9 opens K1 (fail-safe).
Q1 gate: GPIO12 via R1, R2 pull-down.

Fuse block (key side)



Motor bus sense



Notes

Main current path

BT1 + -> F1 40 A -> J1 -> K1 stud A -> K1 stud B
-> TB2 -> B+ of U10-U14 (10 AWG).
B- of U10-U14 -> TB1 star -> J1 -> BT1 -.
The blade fuse block carries only the small
branches; it is not in the 40 A path.
K1 precharge: decide after the bench test
(5 x 470 µF behind a 125 A contactor).

Fuse rule

Every fuse protects the wire after it:
the wire is rated above the fuse.
F1 40 A: 10 AWG. F2 15 A: 14 AWG.
F6 10 A: 16 AWG. F3-F5, F7, F8: 20 AWG
or heavier. See WIRE_LIST.md.



Draft v0.3, not for building the car-facing side until the TBDs close

RideOnESP32 · Can-Am Maverick X3 24 V 4WD

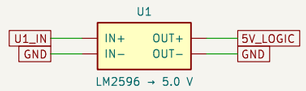
Sheet: /4 Power and safety chain/
File: power.kicad_sch

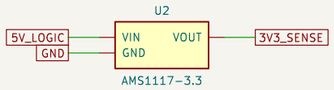
Title: Power and safety chain

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 4/9

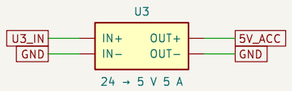
Logic and accessories never share a converter. 3V3_SENSE comes from 5V_LOGIC, not the ESP32's own 3.3 V pin.
Set each buck's output with a meter before connecting any load.

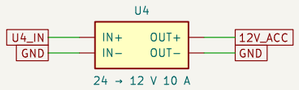
Logic





Accessories





Budget

Rail budget

5V_LOGIC	U1 LM2596	ESP32, RX, halls, U2	~0.5 A	F4 2 A
3V3_SENSE	U2 AMS1117	IBT-2 logic, I2C	~0.1 A	(via F4)
5V_ACC	U3 5 A	DFPlayer, USB, LEDs	≤5 A	F5 3 A
12V_ACC	U4 10-20 A	lights F9, amp F10	≤10 A	F6 10 A

Fuses are on the 24 V input side: 5 A at 5 V is about 1.2 A at 24 V.

Draft v0.3, not for building the car-facing side until the TBDs close

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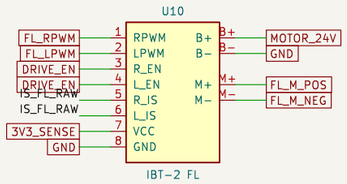
Sheet: /5 Low-voltage rails/
File: rails.kicad_sch

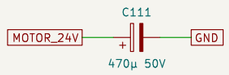
Title: Low-voltage rails

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 5/9

Layer 2: DRIVE_EN and STEER_EN are pulled down at every board, so boot, reset, flashing and crashes leave the drivers off.
IS sense: R_IS + L_IS joined (only one side conducts at a time) -> 4.7k -> ADS1115, 10 nF filter, clamp to 3.3 V.
Variant B (2 x MDD20A) replaces this sheet: see variant_b/.

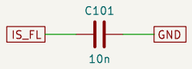
FL (front-left)





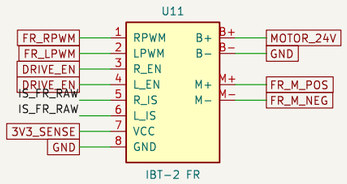


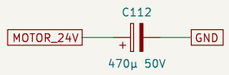






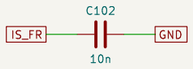
FR (front-right)





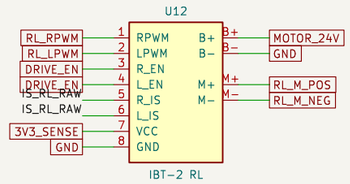


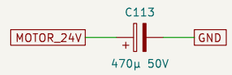






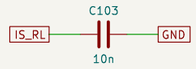
RL (rear-left)





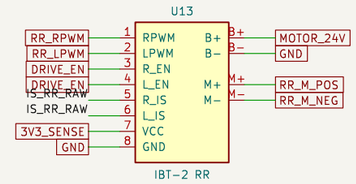


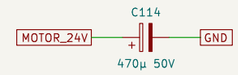






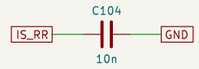
RR (rear-right)





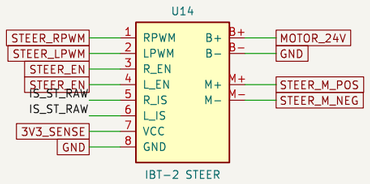


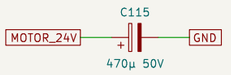




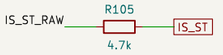


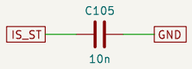
STEER (steering)













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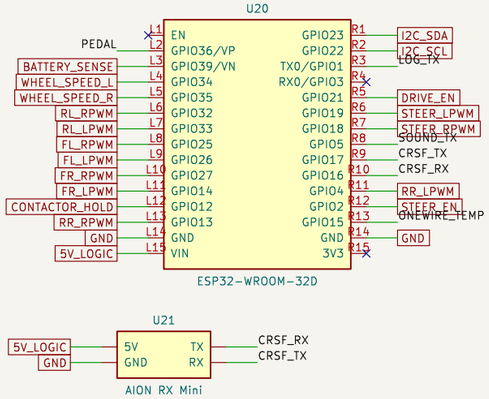
Sheet: /6 Drive, variant A (4 + 1 IBT-2)/
File: drive.kicad_sch

Title: Drive, variant A (4 + 1 IBT-2)

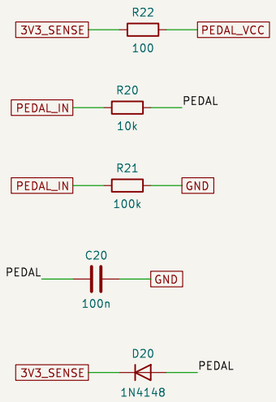
Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 6/9

Pin nets come from include/pins.h; CI keeps pins.h, PINOUT.md and WIRING.md in step and check_netlist.py checks this sheet.
Analog inputs on ADC1 only. GPIO36/39 carry analog only (they glitch when the ADC runs). Kill lines on pins quiet at boot.

ESP32



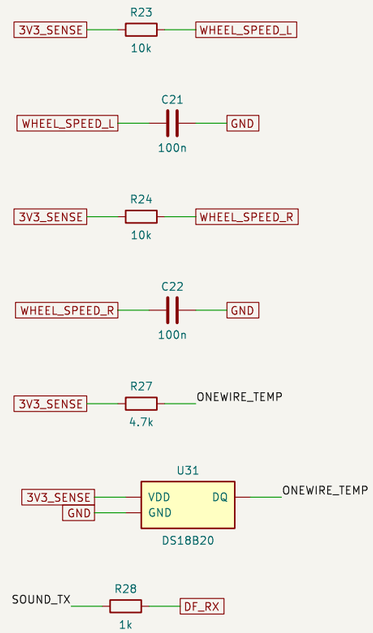
Pedal



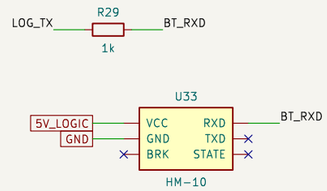
Pedal input

Switch pedal: between V+ and SIG.
Pot: ends on V+ / GND, wiper on SIG.
5 V hall throttle: add a divider (TBD #16).
Shorted high reads as full pedal: open bench check (CLAUDE.md).

Wheel speed, temperature, sound



Bluetooth log, listen-only

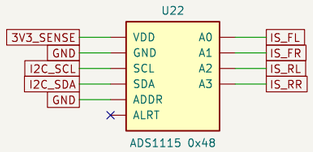


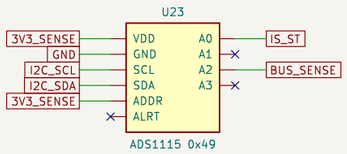
Draft v0.3, not for building the car-facing side until the TBDs close
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Sheet: /7 Controller and inputs/		
File: controller.kicad_sch		
Title: Controller and inputs		
Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 7/9

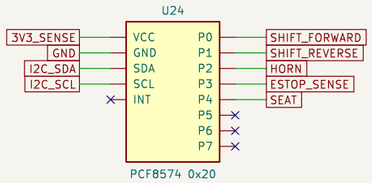
I2C on GPIO22 (SCL) / GPIO23 (SDA) at 3.3 V. Addresses: 0x20 0x36 0x3C 0x40 (+0x70 all-call) 0x48 0x49 0x68.
Off-box I2C (AS5600 on the steering column, OLED on the dash) is the weak spot: keep runs short, twisted with GND (#36).

Current sense ADCs





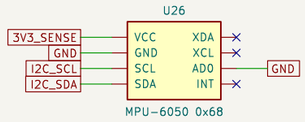
Switches

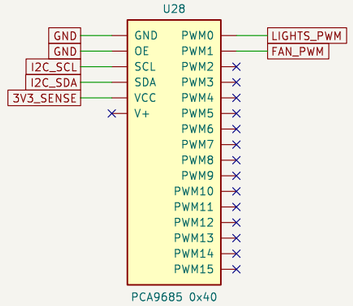


PCF8574 inputs

P0 shift forward, P1 shift reverse, P2 horn,
P3 E-stop sense (NO contact: 0 = pressed),
P4 seat switch. P5–P7 spare.
The E-stop itself cuts K1 in hardware;
P3 only tells the firmware why.

IMU, PWM, pull-ups









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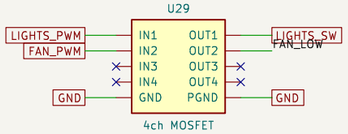
Sheet: /8 I2C bus and switch inputs/
File: i2c_io.kicad_sch

Title: I2C bus and switch inputs

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 8/9

Phase 7. Nothing here is in the 24 Oct order except the inline fuse holders.
Music has its own amp with Bluetooth on that board only; the ESP32 never runs Wi-Fi or Bluetooth.

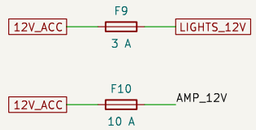
Switched loads

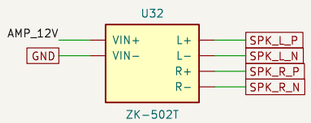






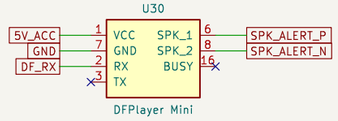
12 V branches







Car sounds



Audio

DFPlayer on 5V_ACC so speaker current stays
off the ESP32 supply. See docs/hardware/AUDIO.md.

Draft v0.3, not for building the car-facing side until the TBDs close

RideOnESP32 · Can-Am Maverick X3 24 V 4WD

Sheet: /9 Accessories: fans, lights, sound/
File: accessories.kicad_sch

Title: Accessories: fans, lights, sound

Size: A3	Date: 2026-10-24	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 9/9